

Question 1:

Given:

```
1. public record Peniche(int length, int draft, int cargoCapacity) {  
2.  
3.     abstract public int fuelConsumptionDuringNavigation( int kilometers );  
4.  
5.     public static boolean setDepartureLocation( int latitude, int longitude  
6. ) {  
7.         return true;  
8.     }  
9.     public void logCruise() {  
10.         System.out.println( "Sailing along the Seine, passing by the Eiffel  
11.         Tower..." );  
12.     }
```

- A. Which method causes a compilation error?
- B. The static setDepartureLocation method.
- C. The abstract fuelConsumptionDuringNavigation method.
- D. Everything is fine.
- E. The instance logCruise method.

Question 2:

Given:

```
1. var cityPopulation = new TreeMap<String, Integer>();
2. cityPopulation.put( "Paris", 2_100_000 );
3. cityPopulation.put( "Marseille", 900_000 );
4. cityPopulation.put( "Lyon", 500_000 );
5. cityPopulation.put( "Lille", 200_000 );
6. cityPopulation.replaceAll( ( city, inhabitants ) -> {
7.     if ( city.startsWith( "L" ) ) {
8.         return inhabitants - 100_000;
9.     } else {
10.        return inhabitants + 100_000;
11.    }
12.} );
13. System.out.println( cityPopulation.values() );
```

What is printed?

- A. [200000, 500000, 900000, 2100000]
- B. [100000, 400000, 1000000, 2200000]
- C. [Lille, Lyon, Marseille, Paris]
- D. An exception is thrown at runtime.
- E. Compilation fails.

Question 3:

Given:

```
1. Stream<String> kings = Stream.of(  
2.     "Charlemagne",  
3.     "Louis IX",  
4.     "Francis I",  
5.     "Louis XIV",  
6.     "Louis XVI" );  
7. Map<Integer, Long> kingsMap = kings.collect(  
8.     Collectors.groupingBy(  
9.         String::length, Collectors.counting()  
10.    ) );  
11. System.out.println( kingsMap.values() );
```

What is printed?

- A. An exception is thrown at runtime.
- B. Compilation fails.
- C. [8, 9, 11]
- D. [1, 3, 1]

Question 4:

Given:

```
1. public static double getPerimeter(Shape shape) throws  
   IllegalArgumentException {  
2.     if (shape instanceof Rectangle r || r.length() > 0) { //n1  
3.         // ...  
4.     } else if (shape instanceof Circle c && c.length() > 0) { //n2  
5.         // ..  
6.     } else {  
7.         // ..  
8.     }  
9. }
```

Which of the following statements is true?

- A. It fails to compile at line n2.
- B. getPerimeter compiles.
- C. It fails to compile at line n1.
- D. An exception is thrown at runtime when executing getPerimeter.

Question 5:

Given:

```
1. public class Test {  
2.     static int x;  
3.  
4.     public static void main( String[] args ) {  
5.         Test t1 = new Test();  
6.         Test t2 = new Test();  
7.     }  
8.  
9.     static { // Line 9  
10.        System.out.print(x);  
11.    }  
12.  
13.    {  
14.        System.out.print(x);  
15.    }  
16. }
```

What is the output?

- A. Compilation fails due to multiple errors
- B. 0
- C. Compilation fails due at line 9
- D. 000
- E. 00

Question 6:

Given:

1. `BinaryOperator<String> longestFrenchWord= BinaryOperator.maxBy(Comparator.comparingInt( String::length ) );`
2. `String unconstitutionally ="Anticonstitutionnellement";`
3. `String hexakosioihexekontahexaphobia="Hexakosioihexekontahexaphobie";`
4. `System.out.println(longestFrenchWord.apply(unconstitutionally,hexakosioihexekontahexaphobia));`

What is printed?

- A. Anticonstitutionnellement
- B. 29
- C. Compilation fails.
- D. Hexakosioihexekontahexaphobie
- E. 25
- F. An exception is thrown at runtime.

Question 7:

Given:

```
1. StringBuilder quoteBuilder = new StringBuilder( "Impossible is not French."
);
2. String quote = new String( quoteBuilder.chars() );
3. quote = quote.concat( " --Napoleon" );
4. System.out.println( quote );
```

What is printed?

- A. Impossible is not French.
- B. Impossible is not French. --Napoleon
- C. --Napoleon
- D. An exception is thrown at runtime.
- E. Compilation fails.

Question 8:

What is true?

- A. `java.lang.System` class has a static field called as `in` of type `PrintStream`
- B. `java.lang.System` class has a static field called as `out` of type `PrintStream`
- C. None of the suggestions
- D. `java.lang.System` class has a static field called as `err` of type `InputStream`
- E. We can use `print()`, `println()`, and `write()` methods in `java.lang.System` class for outputting string

Question 9:

Which of the following will create `LocalDate` instance with day number 300 of the year 2015?

- A. None of these suggestions
- B. `LocalDate.ofYearDay(2015,300);`
- C. `LocalDate.ofYear(2015,300);`
- D. `LocalDate.ofDay(2015,300);`
- E. `new LocalDate(2015,300);`

Question 10:

Given:

```
1. Locale currentLocale = new Locale("fr", "CA", "UNIX");  
2. ResourceBundle introLabels = ResourceBundle.getBundle(  
3.     "ButtonLabel", currentLocale);
```

And given the following list of ResourceBundle:

1. ButtonLabel_fr_CA_UNIX
2. ButtonLabel_fr_CA
3. ButtonLabel_fr
4. ButtonLabel_en_US
5. ButtonLabel_en
6. ButtonLabel

Which ResourceBundle will be selected?

- A. ButtonLabel
- B. ButtonLabel_fr
- C. ButtonLabel_en_US
- D. ButtonLabel_en
- E. ButtonLabel_fr_CA_UNIX

Question 11:

You are asked to create a `ConcurrentHashMap` which allows 15 concurrent writer threads, and an initial capacity of 100.

Which of the following will satisfy the above requirement?

- A. `ConcurrentHashMap map = new ConcurrentHashMap(100, 0.8f, 15);`
- B. `ConcurrentHashMap map = new ConcurrentHashMap(100, 15);`
- C. `HashMap map = new ConcurrentHashMap(100, 15);`
- D. None of the above
- E. `ConcurrentHashMap map = new ConcurrentHashMap(15, 100);`

Question 12:

Given:

```
1. List<Integer> phoneCountryCode = List.of( 1, 32, 33, 34, 44, 49 );
2. <INSERT CODE HERE>
3. phoneCountryCode.forEach( i -> {
4.     if ( onlyFrance.test( i ) ) {
5.         System.out.println( "France international dialing code is
        in the list: " + i );
6.     }
7. } );
```

Which of the following suggestions prints :

"France international dialing code is in the list: 33" ? (Choose 2)

- A. UnaryOperator<Integer> onlyFrance = i -> 33;
- B. IntPredicate onlyFrance = i -> i == 33;
- C. Function<Integer, Boolean> onlyFrance = i -> i == 33;
- D. Predicate<Integer> onlyFrance = i -> i == 33;

Question 13:

Given the following are in the same file:

```
1. public interface IconicFrenchAnimal {
2.     private static int zero(){
3.         return 0;
4.     }
5.
6.     int pawns(boolean usedToWalk);
7. }
8.
9. abstract class Frog implements IconicFrenchAnimal{
10.}
11.
12. abstract class Rooster implements IconicFrenchAnimal{
13.     public abstract short pawns(boolean usedToWalk);
14.}
15.
16. class Snail implements IconicFrenchAnimal{
17.     int pawns(boolean usedToWalk){
18.         return 0;
19.     }
20.}
```

Which types **HAVE** compilation error(s)? (Choose 2)

- A. Rooster
- B. Frog
- C. Snail
- D. IconicFrenchAnimal

Question 14:

Given:

```
1. public class FrenchPoets {  
2.     public static void main( String[] args ) {  
3.         String[] frenchPoet = { "Baudelaire", "Rimbaud", "Verlaine" };  
4.         setHugoPoet( frenchPoet );  
5.         System.out.println( frenchPoet[0] );  
6.     }  
7.  
8.     private static void setHugoPoet( final String[] poets ) {  
9.         if ( poets.length > 0 ) {  
10.             poets[0] = "Hugo";  
11.         }  
12.     }  
13. }
```

What will be printed?

- A. Compilation fails.
- B. Verlaine
- C. Baudelaire
- D. Hugo
- E. An exception is thrown at runtime.

Question 15:

Given:

```
1. public class Test {  
2.     public static void main( String[] args ) { // Line 2  
3.         m( 24 );  
4.     }  
5.  
6.     static void m( int i ) { // Line 6  
7.         System.out.println( i );  
8.         throws new NullPointerException();  
9.     }  
10. }
```

Which of the following modification allows code to compile?

- A. None of these suggestions
- B. Change line 2 to `public static void main( String[] args ) throw Exception {`
- C. Change line 5 to `static void m( int i ) throw RuntimeException {`
- D. No modification is needed
- E. Change line 2 to `public static void main( String[] args ) throw RuntimeException {`

Question 16:

How to instantiate a MonthDay class? Select 2.

- A. `var bastilleDay = MonthDay.of(7, 14);`
- B. `var bastilleDay = MonthDay.of(Month.JULY, 14);`
- C. `var bastilleDay = MonthDay.of(14, 7);`
- D. `var bastilleDay = new MonthDay(7, 14);`
- E. `var bastilleDay = new MonthDay(14, 7);`

Question 17:

Given:

```
1. var croissantList = new ArrayList<>>();
2. var painAuChocolatList = new ArrayList<>>();
3. System.out.print(
4.     ""
5.     +
6.     (croissantList.getClass() == painAuChocolatList.getClass())
7.     +
8.     ", "
9.     + (croissantList == painAuChocolatList)
10.    +
11.    ", "
12.    + (croissantList.equals( painAuChocolatList ))
13.    );
```

What is printed?

- A. true, true, true
- B. false, false, false
- C. true, false, false
- D. true, false, true

Question 18:

Given:

1. `Optional<String> coffee = Optional.of( "coffee" );`
2. `Optional<String> tea = Optional.of( "tea" );`
3. `System.out.println( coffee.or( () -> tea ) );`

What is printed?

- A. `coffee`
- B. `tea`
- C. `Optional[coffee]`
- D. `Optional[tea]`
- E. An exception is thrown at runtime.
- F. Compilation fails.

Question 19:

Given:

```
1. var frenchPlats = new ArrayList();
2.
3. frenchPlats.add( "ratatouille");
4. frenchPlats.add( "bouillabaisse");
5. frenchPlats.add( "coq_au_vin");
6.
7. frenchPlats.set( 3, "carbonade_flamande" );
8.
9. System.out.println(frenchPlats);
```

What will be printed?

- A. [ratatouille, bouillabaisse, coq_au_vin]
- B. An exception is thrown at runtime.
- C. [ratatouille, bouillabaisse, carbonade_flamande]
- D. Compilation fails.
- E. [ratatouille, bouillabaisse, coq_au_vin, carbonade_flamande]

Question 20:

Given:

```
1. interface Changer<T> {  
2.     T change( T t );  
3. }
```

Which of the following lambda expression can be used for change functional method?

- A. `Changer<String> fun = String::length;`
- B. **None of the above**
- C. `Changer<String> fun = () -> "ABC";`
- D. `Changer<String> fun = s -> return s;`
- E. `Changer<String> fun = String::toLowerCase;`

Question 21:

Given:

```
1. public class CharIntSwitch {  
2.     public static void main( String[] args ) {  
3.         char c = 65;  
4.         print( c );  
5.     }  
6.  
7.     static void print( Object o ) {  
8.         switch ( o ) {  
9.             case Integer i -> System.out.println( "int" );  
10.            case Character c -> System.out.println( "char" );  
11.            default -> {  
12.                }  
13.        }  
14.    }  
15.}
```

What is printed?

- A. int
- B. An exception is thrown at runtime.
- C. char
- D. Compilation fails.

Question 22:

You increased some thread priorities while lowering others, but lower-priority threads rarely run.

What concurrency concept describes this phenomenon?

- A. Livelock
- B. Starvation
- C. Race condition
- D. Deadlock

Question 23:

Given:

```
1. public class DataOutputStreamTest {
2.     static final String  dataFile = "invoicedata";
3.     static final double[] prices  = { 19.99, 9.99, 15.99, 3.99, 4.99 };
4.
5.     public static void main( String[] args ) throws FileNotFoundException {
6.
7.         var out = new DataOutputStream( new BufferedOutputStream( new
8. FileOutputStream( dataFile ) ) );
9.         for ( int i = 0; i < prices.length; i++ ) {
10.             out.<INSERT CODE HERE>( prices[i] );
11.         }
12.
13.         var in = new DataInputStream( new BufferedInputStream( new
14. FileInputStream( dataFile ) ) );
15.         double price;
16.         double total = 0.0;
17.         try {
18.             while ( true ) {
19.                 price = in.<INSERT CODE HERE>();
20.                 total += price;
21.             }
22.         } catch ( EOFException e ) {
23.         }
```

What should be inserted in <INSERT CODE HERE> placeholders?

- A. writeInt, readInt
- B. writeByte, readByte
- C. write, read
- D. writeDouble, readDouble

Question 24:

Given:

```
1. public static void main( String[] args ) throws IOException {  
2.     FileWriter outputStream = null;  
3.     try {  
4.         outputStream = new FileWriter( "characterOutput.txt" );  
5.         outputStream.write( 23 );  
6.     } finally {  
7.         if ( outputStream != null ) {  
8.             outputStream.close();  
9.         }  
10.    }  
11. }
```

Choose the right statements. (Choose 2)

- A. It will append the content if the file `characterOutput.txt` exists.
- B. Compilation fails.
- C. It will throw an exception if `characterOutput.txt` is a folder.
- D. It will create the file `characterOutput.txt` if it does not exist.
- E. It will throw an exception if the file `characterOutput.txt` does not exist.

Question 25:

Which of the following is true?

- A. It is optional to use catch for a try block
- B. None of the above
- C. We can't nest few try/catch blocks
- D. We must have a finally clause for a try block

Question 26:

Given:

```
1. LocalDateTime today;  
2. String result;  
3. DateTimeFormatter formatter;  
4.  
5. formatter = DateTimeFormatter.ofPattern( "EEE d MMM yy HH:MM");  
6. today = LocalDateTime.of( 2025,3,15,16,26 );  
7. result = formatter.format(today);  
8. System.out.println("Result: " + result);
```

What is printed?

- A. Result: Sat 15 Mar 25 16:03
- B. Compilation fails.
- C. An exception is thrown at runtime.
- D. Result: Sat 15 Mar 25 16:26

Question 27:

Given:

1. `Stream<String> strings = Stream.of( "1", "2", "3", "4", "5" );`

Which of the following will convert it to a Stream of Integer wrappers?

- A. None of the above
- B. `Stream ints = strings.map( Integer::decode );`
- C. `Stream ints2 = strings.map( Integer::toString );`
- D. `Stream ints = strings.mapToInt( Integer::parseInt );`
- E. `Stream ints = strings.merge( Integer::parseInt );`

Question 28:

In Java Platform Module System, which of the following statements are true?
(Choose 2)

- A. `module-info.java` file can have only one module name
- B. `module-info.java` file must not be empty
- C. `module-info.java` file can be located inside any folder
- D. At compilation, the `module-info.java` becomes a `".properties"` file

Question 29:

Given:

```
1. var frenchSalutations = Stream.of(  
2.     "bonjour",  
3.     "salut",  
4.     "Coucou",  
5.     "enchante",  
6.     "salut" );  
7. var frenchDistinctSalutations = frenchSalutations.peek( System.out::println  
8.     )  
                                     .distinct();
```

What is printed?

- A. An exception is thrown at runtime.
- B. Compilation fails.
- C. bonjour salut Coucou enchante
- D. nothing
- E. bonjour salut Coucou enchante salut

Question 30:

Given:

```
1. public class Cellar {
2.     public static void main( String[] args ) {
3.         HashSet bottles = new HashSet<>();
4.         bottles.add( new Bottle( "Champagne", 75 ) );
5.         bottles.add( new Bottle( "Champagne", 300 ) );
6.         System.out.println( bottles.size() );
7.     }
8. }
9.
10. class Bottle {
11.     String name;
12.     int    cl;
13.
14.     public Bottle( final String name, final int cl ) {
15.         this.name = name;
16.         this.cl = cl;
17.     }
18.
19.     public boolean equals( final Object o ) {
20.         return Objects.equals( name, this.name );
21.     }
22.
23.     public int hashCode() {
24.         return cl;
25.     }
26. }
```

What is printed?

- A. Compilation fails.
- B. 1
- C. 0
- D. 2
- E. An exception is thrown at runtime.

Question 31:

Which of the following method(s) of the Object class can't be overridden?

- A. None of the above
- B. getClass
- C. hashCode
- D. equals
- E. clone

Question 32:

You have to create the Java module AirbusA380Plant named `com.airbus.a380.plant`.

The `com.airbus.a380.plant` module depends on three modules:

`com.airbus.toulouse.assembly`,

`com.airbus.broughton.wings`, and

`com.airbus.hamburg.fuselage`.

How would you define the `module-info.java` related file?

(A)

```
1. module com.airbus.a380.plant{
2.     requires com.airbus.toulouse.assembly;
3.     requires com.airbus.broughton.wings;
4.     requires com.airbus.hamburg.fuselage;
5. }
```

(B)

```
1. module com.airbus.a380.plant{
2.     uses com.airbus.toulouse.assembly;
3.     uses com.airbus.broughton.wings;
4.     uses com.airbus.hamburg.fuselage;
5. }
```

(C)

```
1. module com.airbus.a380.plant{
2.     exports com.airbus.toulouse.assembly;
3.     exports com.airbus.broughton.wings;
4.     exports com.airbus.hamburg.fuselage;
5. }
```

(D)

```
1. module com.airbus.a380.plant{
2.     opens com.airbus.toulouse.assembly;
3.     opens com.airbus.broughton.wings;
4.     opens com.airbus.hamburg.fuselage;
5. }
```

(E)

```
1. module com.airbus.a380.plant{
2.     import com.airbus.toulouse.assembly;
3.     import com.airbus.broughton.wings;
4.     import com.airbus.hamburg.fuselage;
5. }
```


Question 33:

Given:

```
1. double[] arrayEven = { 0, 2, 4, 6, 8 };
2. double[] arrayOdd = { 1, 3, 5, 7, 9 };
3.
4. var compareResult = Arrays.compare( arrayEven, arrayOdd );
5. var mismatchResult = Arrays.mismatch( arrayEven, arrayOdd );
6.
7. System.out.println( compareResult + " " + mismatchResult );
```

What is printed?

- A. 0 0
- B. -1 0
- C. 1 0
- D. It throws an exception at runtime.
- E. Compilation fails.

Question 34:

Given:

```
1. var overseasRegions = List.of(  
2.     "Guadeloupe",  
3.     "Martinique",  
4.     "Guyane",  
5.     "La Réunion",  
6.     "Mayotte"  
7. );  
8. overseasRegions.parallelStream()  
9.     .forEachOrdered( region -> System.out.print( region + ", " )  
    );
```

What will be printed?

- A. An exception is thrown at runtime.
- B. The list of overseas regions in a random ordered.
- C. Always: "Guadeloupe, Martinique, Guyane, La Réunion, Mayotte, "
- D. Compilation fails.

Question 35:

In Java Platform Module System, which of the following is not a module directive?

- A. requires
- B. provides
- C. opens
- D. depends
- E. exports
- F. uses

Question 36:

Given:

```
1. var ordersFile = new File( "orders.csv" );
2. var ordersLines = Files.readAllLines( ordersFile );
3. ordersLines.forEach( (line) -> System.out.println(line));
```

and given the orders.csv file:

```
1. OrderID,Customer,Product,Quantity,Price
2. 1,Kylian Mbappé,Ball,2,25.50
3. 2,Teddy Riner,Black belt,1,15.99
4. 3,Sebastien Loeb,Gloves,1,199.99
5. 4,Antoine Griezmann,Shirt,3,45.00
```

What is printed?

- A. Compilation fails.
- B. The first line of orders.csv.
- C. An exception is thrown at runtime.
- D. The lines of orders.csv.
- E. Nothing.

Question 37:

Given:

```
1. int frenchPopulation = 68_000_000;
2. //France's estimated population growth rate in 2025 is 0.25%
3. long frenchPopulationIn2020 = frenchPopulation * Math.exp( 0.0025 * 5 );
4. System.out.println( String.format(
5.     "Estimated population in 2030: %.1f million",
6.     frenchPopulationIn2020 / 1_000_000 )
7. );
```

What happens?

- A. It throws an exception at runtime.
- B. Compilation fails.
- C. It prints: "Estimated population in 2030: 68.9 million"
- D. It prints: "Estimated population in 2030: 69 million"

Question 38:

Which package contains the `DateTimeFormatter` class?

- A. `java.util.time`
- B. `java.time`
- C. `java.util`
- D. None of above
- E. `java.time.format`

Question 39:

What modifiers are assumed for all interface abstract methods?

- A. protected, final
- B. public, static, and final
- C. public
- D. final, static
- E. public, abstract, static

Question 40:

Given:

1. `Consumer<Integer> cons1 = System.out::print;`
2. `Consumer<Integer> cons2 = cons1.andThen( in -> System.out.print( in * 2 ) );`
- 3.
4. `List<Integer> ins = new ArrayList<>();`
5. `ins.add( 1 );`
6. `ins.add( 2 );`
7. `ins.add( 3 );`
8. `ins.forEach( cons2 );`

What is the output?

- A. 123
- B. 122436
- C. Compilation fails
- D. 123246
- E. An exception is thrown

Question 41:

Given:

```
1. public static void m( int u, final int y ) {  
2.     int x = 10;  
3.     class Cal {  
4.         public void get() {  
5.             // Access here : Line 5  
6.         }  
7.     }  
8.     u = x + 1;  
9. }
```

Which of the following variables can be accessed at line 5?

- A. Only u and x
- B. Only y
- C. None
- D. Only u and y
- E. Only y and x

Question 42:

Given:

```
1. package tgv;
2.
3. public class TGV {
4.     protected String name= "TGV";
5. }
```

And given:

```
1. package tgv.international;
2.
3. import tgv.TGV;
4.
5. public class Eurostar extends TGV {
6.     public static void main( String[] args ) {
7.         TGV eurostar = new Eurostar();
8.         System.out.print(eurostar.name);
9.     }
10. }
```

What will be printed?

- A. An exception is thrown at runtime.
- B. Nothing
- C. Class TGV does not compile.
- D. TGV
- E. Class Eurostar does not compile.

Question 43:

Given:

```
1. public class HiFromFrance {  
2.     public static void main( String[] args ) {  
3.         Object object = "Hello from France!";  
4.         sayHi(object);  
5.     }  
6.  
7.     private static void sayHi( final Object object ) {  
8.         if(object instanceof String String && String.length()>3){  
9.             System.out.println(object);  
10.        }  
11.    }  
12. }
```

What will be printed?

- A. An exception is thrown at runtime.
- B. Hello from France!
- C. Nothing
- D. Compilation fails.

Question 44:

Given:

```
1. public class BreakFor {  
2.     public static void main( String[] args ) {  
3.         stopOnThree:  
4.         for ( var i = 0; i < 3; i++ ) {  
5.             for ( var j = 0; j < 3; j++ ) {  
6.                 if ( i + j > 3 ) {  
7.                     break stopOnThree;  
8.                 }  
9.             }  
10.            System.out.println( "Hello from France!" );  
11.        }  
12.    }  
13. }
```

How many times "Hello from France!" will be printed?

- A. Two
- B. Three
- C. Zero
- D. Compilation fails.
- E. An exception is thrown at runtime.

Question 45:

Given:

```
1. String filePath = "monuments.txt";
2. BufferedReader reader = null;
3. try ( reader =new BufferedReader(new FileReader(filePath ))){
4.     String line;
5.     while ( (line = reader.readLine()) != null ) {
6.         System.out.println( line );
7.     }
8. } catch(IOException e){
9.     System.err.println( "Error reading the file: " + e.getMessage() );
10. }
```

and monuments.txt file is in the right location,
and monuments.txt file contains:

1. Eiffel Tower
2. Arc de Triomphe
3. Notre-Dame de Paris
4. Palace of Versailles
5. Mont Saint-Michel

- A. It throws an exception.
- B. It prints the content of monuments.txt
- C. It runs successfully without printing anything.
- D. Compilation fails.

Question 46:

Given:

1. `List<? extends Number> list = new ArrayList<>(); // Line 1`
2. `list.add( new Integer( 10 ) );`
3. `list.add( new Double( 1.2 ) );`
- 4.
5. `System.out.println( list.get( 0 ) + "" + list.get( 1 ) );`

What is the output?

- A. Compilation fails due to line 1
- B. 101.2
- C. 101
- D. Compilation fails due to multiple errors
- E. 11.2

Question 47:

Given:

```
1. var list = List.of(-10,1,10,100);  
2. var min = list.stream().reduce( Integer.MIN_VALUE,(i,j)->i<j?i:j );  
3. System.out.println(min);
```

What is printed?

- A. Integer.MIN_VALUE
- B. 1
- C. Compilation fails.
- D. An exception is thrown at runtime.
- E. -10

Question 48:

Given:

```
1. import java.time.Period;
2.
3. public class Test {
4.
5.     public static void main( String[] args ) {
6.         System.out.println( Period.ZERO.getUnits() );
7.     }
8.
9. }
```

What is the output?

- A. [Years, Months, Days]
- B. Compilation fails
- C. An exception is thrown
- D. 0
- E. [Days]

Question 49:

Given:

1. `var frenchSpeakingCountries = Stream.of("FR","BE","CA");`
2. `var mix = frenchSpeakingCountries.collect(Collectors.joining("@",
"#","$"));`
3. `System.out.println(mix);`

What is printed?

- A. `@#$FR@#$BE@#$CA@#$`
- B. An exception is thrown at runtime.
- C. Compilation fails.
- D. `#FR@BE@CA$`
- E. `@FR#BE#CA$`

Question 50:

Given:

1. `Function<Integer, Integer> addTwo = i->2+i;`
2. `Function<Integer, Integer> tripleIt = i->3*i;`
3. `Function<Integer, Integer> calculateIt = addTwo.compose( tripleIt );`
4. `System.out.println(calculateIt.apply( 1 ));`

What is printed?

- A. 5
- B. Compilation fails.
- C. 1
- D. An exception is thrown at runtime.
- E. 9